

Markov Chains

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Probability Notes

Random Variables, Expectation, and Key Distributions

1. Random Variables and Expectation

- **Discrete random variables** $\rightarrow$ probability mass function (pmf):

$$p(x) = P(X = x)$$

- **Continuous random variables** $\rightarrow$ probability density function (pdf):

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Expectation ($E[X]$) $\rightarrow$ probability weighted average

$$\text{Discrete} \rightarrow E[X] = \sum x_i p_i$$

$$\text{Continuous} \rightarrow E[X] = \int x f(x) dx$$

2 properties of expectation:

1. **Linearity:** $E[aX + bY] = aE[X] + bE[Y]$
2. **Indicator trick:** for event A , $E[1_A] = P(A)$
(random variable when its value is 1 if it happens, otherwise 0)

2. Variance and Key Properties

Variance: $Var(X) = E[X^2] - (E[X])^2$

Derivation:

$$\begin{aligned}Var(X) &= E[(X - \mu)^2] \quad (\mu \rightarrow \text{mean}) \\&= E[(X - E[X])^2] \\&= E[X^2 - 2XE[X] + (E[X])^2] \quad (\text{Note: } E[X] \text{ is constant}) \\&= E[X^2] - 2E[XE[X]] + (E[X])^2 \\&= E[X^2] - 2E[X]E[X] + (E[X])^2 \\&= E[X^2] - (E[X])^2\end{aligned}$$

Standard Deviation: $\sqrt{Var(X)}$ (By defn)

Properties of Variance

1. **Constant:** $Var(aX + b) = a^2Var(X)$

$$\begin{aligned}Var(aX + b) &= E[((aX + b) - E[aX + b])]^2 \quad (\text{By defn}) \\&= E[((aX + b) - (aE[X] + b))]^2 \\&= E[(aX - aE[X])^2] \\&= E[a^2(X - E[X])^2] \\&= a^2Var(X)\end{aligned}$$

2. **Sum:** $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

$$\begin{aligned}Var(X + Y) &= E[((X + Y) - (E[X] + E[Y]))^2] \\&= E[((X - E[X]) + (Y - E[Y]))^2] \\&= Var(X) + Var(Y) + 2E[XY + E[X]E[Y] - XE[Y] - YE[X]] \\&= Var(X) + Var(Y) + 2\underbrace{(E[XY] - E[X]E[Y])}_{Cov(X,Y) \text{ (By defn)}} \\&= Var(X) + Var(Y) + 2Cov(X, Y)\end{aligned}$$

3. **If X and Y are independent:**

$$Var(X + Y) = Var(X) + Var(Y) \quad \text{i.e. } Cov(X, Y) = 0$$

coz for independent X and Y , $E[XY] = E[X]E[Y]$

3. Some Common Probability Distributions

Discrete Distributions

1) Bernoulli

pmf $\rightarrow P(X = 1) = p$ (success), $P(X = 0) = 1 - p$

- **mean** $\rightarrow E[X] = p$

$$\begin{aligned} E[X] &= \sum x \cdot p(x) \quad (\text{coz discrete}) \\ &= 1 \cdot p + 0 \cdot (1 - p) \\ &= p \end{aligned}$$

- **variance** $\rightarrow Var(X) = p(1 - p)$

$$\begin{aligned} Var(X) &= E[X^2] - (E[X])^2 \\ &= (1^2 \cdot p + 0^2 \cdot (1 - p)) - p^2 \\ &= p - p^2 \\ &= p(1 - p) \end{aligned}$$

2) Binomial

pmf $\rightarrow P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ ($n \rightarrow$ total trials, $k \rightarrow$ success)

- **mean** $\rightarrow E[X] = np$

$$\begin{aligned} E[X] &= \sum_{k=0}^n k \cdot \binom{n}{k} p^k (1 - p)^{n-k} \\ &= \sum_{k=1}^n n \binom{n-1}{k-1} p^k (1 - p)^{n-k} \\ &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1 - p)^{n-k} \\ &= np(p + (1 - p))^{n-1} \\ &= np \end{aligned}$$

- **variance** $\rightarrow Var(X) = np(1 - p)$

Using $E[X^2] - (E[X])^2$, we calculate $E[X(X-1)] + E[X]$:

$$\begin{aligned}
 E[X(X-1)] &= \sum_{k=0}^n k(k-1) \binom{n}{k} p^k (1-p)^{n-k} \\
 &= \sum_{k=2}^n n(n-1) \binom{n-2}{k-2} p^k (1-p)^{n-k} \\
 &= n(n-1)p^2 \sum_{k=2}^n \binom{n-2}{k-2} p^{k-2} (1-p)^{n-k} \\
 &= n(n-1)p^2 (p+1-p)^{n-2} \\
 &= n(n-1)p^2
 \end{aligned}$$

So, $Var(X) = n(n-1)p^2 + np - (np)^2 = np(1-p)$

OR Use indicator trick!!!

$X = I_1 + I_2 + \dots + I_n$ where I_i represents event for i^{th} trial.

$$E[X] = \sum_{i=1}^n E[I_i] = np \quad (\text{coz } E[I_i] = p \text{ for each } 1 \leq i \leq n)$$

$$Var(X) = \sum_{i=1}^n Var(I_i) = np(1-p) \quad (\text{coz } Var(I_i) = p(1-p) \text{ for each } 1 \leq i \leq n)$$

3) Poisson

Binomial to Poisson $\rightarrow$ imagine taking n trials and multiplying them toward infinity, while prob. p of each trial shrinks towards zero, such that avg rate of success ($n \times p$) remains const. $\rightarrow$ this constant is λ (rate).

pmf $\rightarrow P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ for $k \geq 0$

Binomial dist. $\rightarrow$ Binomial thm. Poisson dist. $\rightarrow$ Taylor series (expansion of e^x):
 $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

- **mean** $\rightarrow E[X] = \lambda$

$$E[X] = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda} \lambda^k}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

- **variance** $\rightarrow Var(X) = \lambda$

$$Var(X) = E[X^2] - (E[X])^2 \quad (E[X^2] \text{ computed using } E[X(X-1)] + E[X])$$

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k!} = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2$$

$$Var(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

For Poisson's dist. both mean and variance are λ .

Continuous Distributions

Total area = 1 ofc

1) Uniform

pdf $\rightarrow f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$ ($a \rightarrow$ lower limit, $b \rightarrow$ upper limit)

- **mean** $\rightarrow E[X] = \frac{a+b}{2}$

$$\int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{2(b-a)} [x^2]_a^b = \frac{1}{2(b-a)} (b^2 - a^2) = \frac{a+b}{2}$$

- **variance** $\rightarrow Var(X) = \frac{(b-a)^2}{12}$

$$Var(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{3(b-a)} [x^3]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + a^2 + ab}{3}$$

$$\begin{aligned} Var(X) &= \frac{b^2 + a^2 + ab}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{b^2 + a^2 + ab}{3} - \frac{a^2 + b^2 + 2ab}{4} \\ &= \frac{4a^2 + 4b^2 + 4ab - 3a^2 - 3b^2 - 6ab}{12} \\ &= \frac{a^2 + b^2 - 2ab}{12} = \frac{(b-a)^2}{12} \end{aligned}$$

2) Normal (or Gaussian)

Here mean (μ) and variance (σ^2) are the parameters for defining the distribution. (Bell curve)

$$\text{pdf} \rightarrow f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **mean** $\rightarrow E[X] = \mu$

$$\begin{aligned}
 E[X] &= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
 &\text{put } z = \frac{x-\mu}{\sigma} \text{ i.e. } x = z\sigma + \mu, dx = \sigma dz \\
 &= \int_{-\infty}^{\infty} \frac{z\sigma + \mu}{\sqrt{2\pi\sigma^2}} e^{-\frac{z^2}{2}} \sigma dz \\
 &= \frac{\sigma}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz}_{\text{odd fn} \rightarrow 0} + \mu \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{Standard Normal dist, AUC}=1} \\
 &= 0 + \mu(1) = \mu
 \end{aligned}$$

- **variance** $\rightarrow Var(X) = \sigma^2$

$$\begin{aligned}
 Var(X) &= E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
 &\text{put } z = \frac{x-\mu}{\sigma} \text{ i.e. } x = z\sigma + \mu, dx = \sigma dz \\
 &= \int_{-\infty}^{\infty} \frac{z^2\sigma^2}{\sqrt{2\pi\sigma^2}} e^{-\frac{z^2}{2}} \sigma dz \\
 &= \sigma^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} z^2 e^{-\frac{z^2}{2}} dz
 \end{aligned}$$

Use by parts:

$$\begin{aligned}
 &= \sigma^2 \frac{1}{\sqrt{2\pi}} \left(\left[-ze^{-z^2/2} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \right) \\
 &= \sigma^2 \left(0 + \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{Standard Normal Dist.}} \right) \\
 &= \sigma^2
 \end{aligned}$$

4. Independence, Covariance, and Conditional Probability

Independence and Covariance

- A and B are independent if $P(A \cap B) = P(A)P(B)$.

- Random variables X, Y are independent if their joint distribution factors:

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

(Marginal probability dist $X = x$ and ignore Y)

- For independent X & $Y \rightarrow E[XY] = E[X]E[Y]$ and $Var(X + Y) = Var(X) + Var(Y)$.

Covariance measures dirⁿ of dependence

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

The correlation coefficient normalises to $[-1, 1]$:

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

- $\rho = 0 \rightarrow$ uncorrelated but not necessarily independent.
- $\rho = 1 \rightarrow$ perfect +ve linear relationship (converse is true tho).
- $\rho = -1 \rightarrow$ perfect -ve linear relationship.

Conditional Probability and Bayes' Theorem

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' Thm:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Introduction to Markov Chains

Stochastic Process

A stochastic process is simply a collection of random variables, usually indexed by time: $\{X_t, t \in T\}$

where, t is time (discrete or continuous)

X_t is the state of the process at time t

The set of all possible values that X_t can take is called the **state space**.

Markov Property (Memorylessness)

A sequence of random variables is a Markov Chain if it possesses the Markov Property.

Mathematically:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i)$$

i.e. The probability of moving to state j tomorrow (X_{n+1}), given the entire history of the system from day zero, is exactly the same as the probability of moving to state j given only where we are today ($X_n = i$).

One-Step Transition Probabilities (P_{ij})

The probability of transitioning from state i to state j in exactly one time step is P_{ij} .

$$P_{ij} = P(X_{n+1} = j \mid X_n = i)$$

These P_{ij} 's should satisfy two rules:

1. Non-negativity: $P_{ij} \geq 0$
2. Row sums to 1: $\sum_{j=0}^{\infty} P_{ij} = 1 \quad \forall i$

Transition Probability Matrix (P)

All these individual transition probabilities when packaged into a single square matrix (P)

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \dots \\ P_{10} & P_{11} & \dots & \\ \vdots & \vdots & & \\ P_{i0} & P_{i1} & \dots & \\ \vdots & \vdots & & \end{bmatrix}$$

Chapman - Kolmogorov Equations

Core Problem: We know P_{ij} which is probability of going from i to j in exactly 1 step.

But what if we want to know the probability of going from i to j in exactly n steps?
We denote this as $P_{ij}^{(n)}$.

Equation:

To figure out the probability of going from i to j in $n + m$ steps, you have to account for every possible intermediate state k .

Formally:

$$P_{ij}^{(n+m)} = \sum_{k=0}^{\infty} P_{ik}^{(n)} P_{kj}^{(m)}$$

If we let $P^{(n)}$ denote the matrix of n -step transition probabilities $P_{ij}^{(n)}$, then

$$P^{(n+m)} = P^{(n)} \cdot P^{(m)}$$

$\hookrightarrow$ matrix multiplication

Examples

From Introduction to Probability Models by Sheldon Ross:

Example 4.10. An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. If initially both balls are red, find the probability that the fifth ball selected is red.

Solution:

states:

state 0 $\rightarrow$ 0 Red 2 Blue

state 1 $\rightarrow$ 1 Red 1 Blue

state 2 $\rightarrow$ 2 Red 0 Blue

Let X_n be the no. of red balls in urn after n th selection and subsequent replacement.

Transition Matrix:

$$P = \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix}$$

P(fifth selection is red)

$$\begin{aligned} &= \sum_{i=0}^2 P(\text{fifth selection is red} \mid X_4 = i) P(X_4 = i \mid X_0 = 2) \\ &= (0)P_{2,0}^{(4)} + (0.5)P_{2,1}^{(4)} + (1)P_{2,2}^{(4)} \\ &= 0.7048 \quad (\text{after carrying matrix multiplications}) \end{aligned}$$

Example 4.11. Suppose that balls are successively distributed among 8 urns, with each ball being equally likely to be put in any of these urns. What is the probability that there will be exactly 3 nonempty urns after 9 balls have been distributed?

Solution:

→ X_n be the no of nonempty urns after n balls have been distributed.

states: 0 to 8

$$P_{i,i} = i/8$$

$$P_{i,i+1} = 1 - P_{i,i}$$

We want to find $P_{0,3}^{(9)}$.

$$P_{0,3}^{(9)} = P_{0,1}^{(1)} \cdot P_{1,3}^{(8)} = 1 \cdot P_{1,3}^{(8)} = P_{1,3}^{(8)}$$

To build a transition matrix, we have states 1, 2, 3, 4 (in fact, ≥ 4 nonempty urns).

$$P = \begin{bmatrix} 1/8 & 7/8 & 0 & 0 \\ 0 & 2/8 & 6/8 & 0 \\ 0 & 0 & 3/8 & 5/8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

we need $P_{1,3}^{(8)}$.

After Matrix multiplications

$$P_{1,3}^{(8)} = 0.00756$$

Example 4.12. In a sequence of independent flips of a fair coin, let N denote the number of flips until there is a run of three consecutive heads. Find

(a) $P(N \leq 8)$ and

(b) $P(N = 8)$.

Solution:

→ Markov chain, as we are currently on a run of i consecutive heads.

states: 0, 1, 2, 3 (3 → 3 consecutive heads have already occurred)

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We want:

a) $P_{0,3}^{(8)}$

b) $P_{0,2}^{(7)} P_{2,3}^{(1)}$

Calculate P^7 and P^8 to get the answer.

State Classification

- State j is said to be accessible from state i if $P_{ij}^n > 0$ for some $n \geq 0$.

Proposition: State j is accessible from state i if and only if, starting in i , it is possible that the process will ever enter state j .

Proof: If j is not accessible from i , then

$$\begin{aligned}
 P\{\text{ever be in } j \mid \text{start in } i\} &= P\left\{\bigcup_{n=0}^{\infty} \{X_n = j\} \mid X_0 = i\right\} \\
 &\leq \sum_{n=0}^{\infty} P\{X_n = j \mid X_0 = i\} \\
 &= \sum_{n=0}^{\infty} P_{ij}^n \\
 &= 0
 \end{aligned}$$

Two states i and j that are accessible to each other are said to communicate, denoted by $i \leftrightarrow j$.

The relation of communication follows three rules:

1. A state always communicates with itself. (Type of reflexive property)
2. It works both ways. If state i communicates with state j , then state j automatically communicates with state i . (Type of symmetric property)
3. It creates a chain reaction. If state i can reach state j , and state j can reach state k , then state i can definitely reach state k . The math equation in the image simply proves this: if it takes n steps to get to j and m steps to get to k , it will take $n + m$ steps to get from i to k . (Type of transitive property)

Classes: States that all communicate with one another are grouped into a **class**. Two classes are either exactly the same or completely separate (disjoint).

Irreducible: If every single state can reach every other state, the Markov chain is called **irreducible**.

Absorbing State: A state from which no other state is accessible. State i is absorbing if $P_{ii} = 1$.

Let f_i denote the probability that, starting in state i , the process will *ever* reenter state i .

Recurrent State: A state i is recurrent if $f_i = 1$. If a process starts in a recurrent state, it will reenter that state with 100% probability. After each reentry, it is like the process again starts over so if a state is recurrent then it will reenter infinitely often.

Transient State: A state i is transient if $f_i < 1$. Each time the process enters a transient state, there is a positive probability $(1 - f_i)$ that it will *never* enter it again. The probability of being in state i for exactly n time periods is $f_i^{n-1}(1 - f_i)$ for $n \geq 1$.

- The total number of time periods spent in a transient state follows a geometric distribution with a finite mean of $1/(1 - f_i)$.

Corollary:

If state i is recurrent, and state i communicates with state j , then state j is also recurrent.

Proof:

To prove this, we first note that since state i communicates with state j , there exist integers k and m such that $P_{ij}^k > 0$ and $P_{ji}^m > 0$.

Now, for any integer n :

$$P_{jj}^{m+n+k} \geq P_{ji}^m P_{ii}^n P_{ij}^k$$

This inequality holds because the left side represents the total probability of going from state j to state j in $m + n + k$ steps. The right side represents the probability of going from j to j in $m + n + k$ steps via a specific, restricted path:

1. Going from j to i in exactly m steps.
2. Then going from i to i in an additional n steps.
3. Then going from i to j in an additional k steps.

From the preceding inequality, we obtain, by summing over n , that:

$$\sum_{n=1}^{\infty} P_{jj}^{m+n+k} \geq P_{ji}^m P_{ij}^k \sum_{n=1}^{\infty} P_{ii}^n = \infty$$

This evaluates to infinity because $P_{ji}^m P_{ij}^k > 0$ (as established by the definition of communication), and the summation $\sum_{n=1}^{\infty} P_{ii}^n$ is infinite because state i is known to be recurrent.

Example 4.18. Consider the Markov chain having states 0, 1, 2, 3, 4 and

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 & 0 \\ 1/4 & 1/4 & 0 & 0 & 1/2 \end{bmatrix}$$

Determine the recurrent state.

Solution: This chain consists of the three classes $\{0, 1\}$, $\{2, 3\}$, and $\{4\}$. The first two classes are recurrent and the third transient.

Analysing States in a Random Walk

Consider a Markov chain whose state space consists of the integers $i = 0, \pm 1, \pm 2, \dots$, and has transition probabilities given by $P_{i,i+1} = p = 1 - P_{i,i-1}$, $i = 0, \pm 1, \pm 2, \dots$ where $0 < p < 1$.

Since all states clearly communicate, the states are either all transient or all recurrent. So we have to determine whether the summation $\sum_{n=1}^{\infty} P_{00}^n$ is finite or infinite.

It is impossible to reenter the original state exactly in an odd number of moves. It is only possible to reenter the original state in an even number of moves if half the times we go right and half the times left.

$$\begin{aligned} P_{00}^{2n} &= \binom{2n}{n} p^n (1-p)^n \\ &= \frac{(2n)!}{n!n!} (p(1-p))^n, \quad n = 1, 2, 3, \dots \end{aligned}$$

By using an approximation, due to Stirling, which asserts that:

$$n! \sim n^{n+1/2} e^{-n} \sqrt{2\pi} \quad (4.3)$$

where we say that $a_n \sim b_n$ when $\lim_{n \rightarrow \infty} a_n/b_n = 1$, we obtain:

$$\binom{2n}{n} \sim \frac{(2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}}{n^{2n+1} e^{-2n} (2\pi)} = \frac{2^{2n}}{\sqrt{n\pi}}$$

Hence,

$$P_{00}^{2n} \sim \frac{(4p(1-p))^n}{\sqrt{\pi n}}$$

Now it is easy to verify, for positive a_n, b_n , that if $a_n \sim b_n$, then $\sum_n a_n < \infty$ if and only if $\sum_n b_n < \infty$. Hence, $\sum_{n=1}^{\infty} P_{00}^{2n}$ will converge if and only if

$$\sum_{n=1}^{\infty} \frac{(4p(1-p))^n}{\sqrt{\pi n}}$$

does. However, $4p(1-p) \leq 1$ with equality holding if and only if $p = 1/2$. Hence, $\sum_{n=1}^{\infty} P_{00}^{2n} = \infty$ if and only if $p = 1/2$. Thus, the chain is recurrent when $p = 1/2$ and transient if $p \neq 1/2$.

Therefore for $p = 1/2$ all the states are recurrent, elsewhere transient.

Also if we take a 2-dimensional walk i.e. 4 states (right, left, up, down), each having a probability of $1/4$, then here too all the states are recurrent.

Interestingly, whereas the symmetric random walks in one and two dimensions are both recurrent, all higher-dimensional symmetric random walks turn out to be transient.

For a 1-dimensional walk we can prove this in another way too:

Let $\beta = P(\text{ever return to } 0)$

$$\beta = P(\text{ever return to } 0 \mid X_1 = 1)p + P(\text{ever return to } 0 \mid X_1 = -1)(1 - p)$$

Now, let α be the probability that the Markov chain currently in state i will ever enter state $i - 1$, for any i .

$$\begin{aligned}\alpha &= P(\text{ever return} \mid X_1 = 1, X_2 = 0)(1 - p) + P(\text{ever return} \mid X_1 = 1, X_2 = 2)p \\ &= 1 - p + P(\text{ever return} \mid X_1 = 1, X_2 = 2)p \\ &= 1 - p + p\alpha^2\end{aligned}$$

The two roots of this equation are $\alpha = 1$ and $\alpha = (1 - p)/p$.

In case of a Symmetric Random Walk, $p = 1/2$, we get $\alpha = 1$, proving that it is recurrent.

Suppose now that $p > 1/2$, therefore:

$$\beta = \alpha p + 1 - p$$

Because the random walk is transient in this case we know that $\beta < 1$, showing that $\alpha \neq 1$.

Therefore, $\alpha = (1 - p)/p$,
so, $\beta = 2(1 - p)$, for $p > 1/2$.

Similarly, when $p < 1/2$ we can show that $\beta = 2p$.

Thus in general,

$$P\{\text{ever return to } 0\} = 2 \min(p, 1 - p)$$

Long-Run Proportions and Limiting Probabilities

Proposition: If i is recurrent and i communicates with j , then $f_{ij} = 1$.

In short: If state i loops infinitely, and state j can reach state i and come back, then those infinite loops at i drag j into looping infinitely too.

Let $N_j = \min\{n > 0 : X_n = j\}$.

If state j is recurrent, let m_j denote the expected number of transitions that it takes the Markov chain when starting in state j to return to that state.

$$m_j = E[N_j \mid X_0 = j]$$

The recurrent state j is **positive recurrent** if $m_j < \infty$ and it is **null recurrent** if $m_j = \infty$.

Proposition: If the Markov chain is irreducible and recurrent, then for any initial state:

$$\pi_j = \frac{1}{m_j}$$

Proof:

Suppose the process starts in state i .

- Let T_1 = the number of steps to reach state j for the *first* time.
- Let $T_2, T_3, \dots, T_n$ = the number of steps between subsequent visits to state j .

The n -th visit to state j happens at time $T_1 + T_2 + \dots + T_n$.

Therefore, the proportion of time spent in state j over the long run is n visits divided by the total time it took to get there:

$$\pi_j = \lim_{n \rightarrow \infty} \frac{n}{\sum_{i=1}^n T_i}$$

$$\pi_j = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n} \sum_{i=1}^n T_i}$$

$$\pi_j = \lim_{n \rightarrow \infty} \frac{1}{\frac{T_1}{n} + \frac{T_2 + \dots + T_n}{n}}$$

As the number of visits (n) approaches infinity:

- $\lim_{n \rightarrow \infty} \frac{T_1}{n} = 0$ (Because the time to first reach j is finite, dividing it by infinity yields zero).
- $\lim_{n \rightarrow \infty} \frac{T_2 + \dots + T_n}{n} = m_j$ (By the Strong Law of Large Numbers, the average of these repeated, identical return times converges exactly to their expected mean, m_j).

Therefore,

$$\pi_j = \frac{1}{0 + m_j} = \frac{1}{m_j}$$

Imagine a Markov chain running for an infinitely long time. π_j is the exact fraction (or percentage) of that total time the system spends sitting in state j .

If you let the Markov chain run for a very long time, π_j is the probability that you will find the system in state j at some random point far in the future, regardless of where it originally started.

Proposition: If i is positive recurrent and $i \leftrightarrow j$ then j is positive recurrent.

Ergodic Theorem: Consider an irreducible Markov chain. If the chain is positive recurrent, then the long-run proportions are the unique solution of the equations:

$$\pi_j = \sum_i \pi_i P_{ij}, \quad j \geq 0, \quad \sum_j \pi_j = 1$$

Moreover, if there is no solution to the preceding linear equations, then the Markov chain is either transient or null recurrent and all $\pi_j = 0$.

Convergence: As $n \rightarrow \infty$, the transition probabilities P_{ij}^n of certain Markov chains converge to a specific value that is independent of the initial state i .

Periodic Chains: A Markov chain is considered periodic if it can only return to a state in a multiple of $d > 1$ steps. Periodic chains do not have limiting probabilities, although they can still have long-run proportions (e.g., a two-state chain that continually alternates between states 0 and 1).

Aperiodic Chains: An irreducible chain that is not periodic is called aperiodic. For these chains, limiting probabilities always exist, do not depend on the initial state, and are equal to the long-run proportions (π_j).

Equivalence to Long-Run Proportions: Because the set $\{\alpha_j, j \geq 0\}$ satisfies the exact same equations for which the long-run proportions $\{\pi_j, j \geq 0\}$ are the unique solution, it proves that $\alpha_j = \pi_j$.

Ergodic Chains: A Markov chain that is irreducible, positive recurrent, and aperiodic is called **ergodic**.

Markov Chains (Python Implementation)

Markov Chain Object-Oriented Implementation

Below is the Python implementation of a Markov Chain, consolidating core functionalities of what learnt till now such as n-step transitions, calculating stationary distributions, identifying absorbing states, and determining communicating classes using the Floyd-Warshall algorithm.

```
1 import numpy as np
2
3 class MarkovChain:
4     def __init__(self, states, transition_matrix):
5         # Initializing our Markov Chain.
6
7         self.states = states
8         self.transition_matrix = np.array(transition_matrix)
9
10        # Validate the matrix by its 2 basic rules
11        assert self.transition_matrix.shape[0] == self.
transition_matrix.shape[1], "Matrix must be a square."
12        for row in self.transition_matrix:
13            assert np.isclose(sum(row), 1.0), f"Row probabilities must
sum to 1. Found: {sum(row)}"
14
15        def get_n_step_transition(self, n):
16            # To Calculate the n-step transition matrix using matrix
exponentiation.
17            return np.linalg.matrix_power(self.transition_matrix, n)
18
19        def calculate_stationary_distribution(self):
20            # To Calculate the stationary distribution (pi) where  $\pi \cdot P = \pi$ 
.
21            # This is equivalent to finding the left eigenvector associated
with eigenvalue 1.
22
23            # First Transpose the matrix to find left eigenvectors using
standard right eigenvector functions
24            eigenvalues, eigenvectors = np.linalg.eig(self.
transition_matrix.T)
25            # Then find the index of the eigenvalue that is closest to 1
eigen_1_index = np.argmin(np.abs(eigenvalues - 1.0))
26            # Then extract the corresponding eigenvector
stationary_vector = eigenvectors[:, eigen_1_index].real
27            # And finally normalize the vector so probabilities sum to 1
stationary_distribution = stationary_vector / np.sum(
stationary_vector)
28
29            return stationary_distribution
30
31
32        def display_stationary(self):
33            # To print the stationary distribution in a readable format
pi = self.calculate_stationary_distribution()
```

```

37     print("\n>>> Stationary Distribution <<<")
38     for state, prob in zip(self.states, pi):
39         print(f"State {state}: {prob:.4f} ({prob*100:.2f}%)")
40
41     def get_absorbing_states(self):
42         # To get absorbing states
43         absorbing = []
44         for i, state in enumerate(self.states):
45             if np.isclose(self.transition_matrix[i, i], 1.0):
46                 absorbing.append(state)
47         return absorbing
48
49     def simulate_path(self, start_state, num_steps):
50         # To simulate a random walk through the Markov Chain.
51
52         # The starting state should be a valid state ofc
53         if start_state not in self.states:
54             raise ValueError(f"State '{start_state}' not found in state
55 space.")
56
57         current_state_idx = self.states.index(start_state)
58         path = [start_state]
59
60         for i in range(num_steps):
61             next_state_idx = np.random.choice(
62                 len(self.states),
63                 p=self.transition_matrix[current_state_idx]
64             )
65             path.append(self.states[next_state_idx])
66             current_state_idx = next_state_idx
67
68         return path
69
70     def get_communicating_classes(self):
71         # To identify communicating classes (Strongly Connected
72         # Components) using reachability to help classify states as transient
73         # or recurrent.
74
75         n = len(self.states)
76         # First we will need a Boolean reachability matrix (which
77         # answers, can i reach j directly?)
78         reachability = np.zeros((n, n), dtype=bool)
79         np.fill_diagonal(reachability, True)
80         reachability = reachability | (self.transition_matrix > 0)
81
82         # Using Floyd-Warshall algorithm
83         for k in range(n):
84             for i in range(n):
85                 for j in range(n):
86                     reachability[i, j] = reachability[i, j] or (
87                         reachability[i, k] and reachability[k, j])
88
89         # Group states into communicating classes
90         visited = set()
91         classes = []
92         for i in range(n):
93             if i not in visited:

```

```

89         # A class includes all states 'j' where 'i' can reach '
j' AND 'j' can reach 'i'
90         current_class_indices = [j for j in range(n) if
reachability[i, j] and reachability[j, i]]
91         classes.append([self.states[idx] for idx in
current_class_indices])
92         visited.update(current_class_indices)
93
94     return classes
95
96
97 if __name__ == "__main__":
98     # EXAMPLE 1: Weather Model
99
100     # Defining the states
101     weather_states = ['Sunny', 'Cloudy', 'Rainy']
102
103     # Defining the Transition Matrix (P)
104     # Row 1 (Sunny): 60% chance tomorrow is Sunny, 30% Cloudy, 10%
Rainy
105     # Row 2 (Cloudy): 40% Sunny, 40% Cloudy, 20% Rainy
106     # Row 3 (Rainy): 20% Sunny, 50% Cloudy, 30% Rainy
107     P = [[0.6, 0.3, 0.1],
108           [0.4, 0.4, 0.2],
109           [0.2, 0.5, 0.3]]
110
111     # Initializing the chain
112     mc = MarkovChain(weather_states, P)
113
114     # Q. What happens after 5 days?
115     print("\n>>> 5-Step Transition Matrix (P^5) <<<<")
116     print(np.round(mc.get_n_step_transition(5), 4))
117
118     # Q. Calculate the Stationary Distribution
119     mc.display_stationary()
120
121     # Q. Simulate 10 days of weather starting from a Sunny day
122     print("\n>>> 10-Day Weather Simulation <<<")
123     weather_path = mc.simulate_path('Sunny', 10)
124     print(" -> ".join(weather_path))
125
126
127     # EXAMPLE 2: The Gambler's Ruin (Absorbing States & Classes)
128     print("Example 2: Gambler's Ruin ($0 to $3)")
129
130     gambler_states = ['$0', '$1', '$2', '$3']
131     # p = 0.5 (fair coin flip to win or lose $1)
132     gambler_P = [
133         [1.0, 0.0, 0.0, 0.0],
134         [0.5, 0.0, 0.5, 0.0],
135         [0.0, 0.0, 0.0, 1.0],
136         [0.5, 0.0, 0.5, 0.0]
137     ]
138
139     gambler_mc = MarkovChain(gambler_states, gambler_P)
140
141     print("\n>>> Absorbing States <<<")
142     print(gambler_mc.get_absorbing_states())

```

```

143
144 print("\n>>> Communicating Classes <<<")
145 classes = gambler_mc.get_communicating_classes()
146 for i, c in enumerate(classes):
147     # A class with only 1 state that is absorbing is recurrent.
148     # A class that can reach an absorbing state but isn't absorbing
    itself is transient.
149     print(f"Class {i+1}: {c}")
150
151 print("\n>>> Gambler Simulation (Starting at $1) <<<")
152 gambler_path = gambler_mc.simulate_path('$1', 7)
153 print(" -> ".join(gambler_path))

```